Ivachev Fedor (费杰)

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Self-Introduction

- I am a PhD candidate in Computer Science at Tsinghua University, focusing on **Human-Computer Interaction** — specifically context-aware conversational systems based on real-world scenarios.
- I have accumulated engineering experience at international tech companies such as Huawei and Samsung, with full-stack capabilities ranging from low-level algorithm development to the deployment of large language model applications, covering **Computer Vision**, **Mixed Reality**, and **AIoT**.
- Trilingual (English/Chinese/Russian), I am committed to promoting innovative applications of AI in diverse scenarios.

Work Experience

Tsinghua University, Beijing (Teaching Assistant, 2021/01 - 2022/01)

- Assisted lead instructor in facilitating the Human-Computer Interaction course.
- Helped and taught both international and local students.
- Developed educational tutorials and provided assistance for student group projects.

Huawei Technologies Co., Ltd, Moscow (Software Engineer, 2019/02 - 2019/08)

- Worked on low-level graphic profiling tools using **C++**.
- Retrieved data from integrated GPU counters for more precise draw call timing.
- These tools were used by graphic optimization and QA teams.
- Coordinated with developers across Russia and China.

Samsung RC, Ltd, Moscow (Intern, Programmer, 2018/06 - 2018/09)

- Developed computer vision algorithms for the 3D Avatar team using **C++** and **Python**.
- Modified OpenCV source code to correct camera distortion and implemented facial physical landmark detection.
- The solution was later integrated into the operating system release.

Education

- PhD in Computer Science (2021/09-2026/07), Tsinghua University, China
- Master's in Computer Science (2019/09-2021/06), GPA 3.9, Tsinghua University, China
- Bachelor's in Computer Science (2015/09-2019/06), Lomonosov Moscow State University, Russia

Projects & Technology Stack

- **OpenAI API + Azure AI + Huawei Mobile Services** — AI real-world agent: an LLM-based agent that uses multimodal information from sensors to build comprehensive human portraits and manage real-world contextual conversations.
- **Unity C# + IoT platforms** — NUIX-Studio APP: a platform to test AIoT environments in VR. Based on Microsoft MRTK, OpenHAB, and HomeAssistant (v1), it lets users extend real-world IoT devices with VR functionality. Version 2 uses Oculus SDK and includes a UI for prototyping IoT interactions without programming knowledge.
- **Blender** — TikTok effects: I created 10 advanced AR effects for TikTok (Douyin), used over 3 million times.

Qualifications and Awards

- Tsinghua University-Goertek Innovation Proposal Competition — Outstanding Mentor Award (2024)
- TCL RayNeo AI Challenge — 3rd Place (2024)
- Huawei Certification: HCIA-AI (2020)
- MSU Higher School of Business — Best Startup 2018: AI Vehicle Damage Assessment (2018)
- Certificate: 2-week Web App Development Practice, Samsung RC, Moscow (2017)
- Yandex.Money Hackathon — Most Technological Bot (2016)
- National and First-level Olympiads in Mathematics (2015, 2012); National Olympiad in Competitive Programming (2014)

Publications

- Modification of Algorithm for inverting matrices with elements from ring of scalar differential operators. CMC MSU MAKS PRESS 2019: 88 (in Russian)
- LinguaReality: Exploring Context-Aware Language Learning through Conversational AI. UIST 2026 (CCF A; under submission)

Misc. Information

- Languages: English (C2, IELTS 7.5), Chinese (B2, HSK3 295/300), Russian (native)

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